NPWS
An tSeirbhís Páirceanna Náisiúnta agus Fiadhúlra
National Parks and Wildlife Service

INSTALLING STONE DAMS

A STEP-BY-STEP FIELD INSTALLATION & QUALITY GUIDE

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Wicklow Mountains National Park

Overview

> Stone dams are engineered structures installed in peatlands in large ditches and gullies that have eroded down to the mineral layer. Their primary purpose is to slow water flow, trap sediment, raise the drain bed level over time, and prevent the peat walls from drying out, cracking, and collapsing.

1. Site & Location Assessment

Proper positioning and density are critical for ensuring that stone dams perform effectively and withstand hydrologic forces without causing side-scouring of banks.

Ideal Topography & Substrate

- **Substrate:** Stone dams area most effective in gullies that have eroded down to the mineral layer or shallow peat.
- **Slope Gradient:** Best suited for flatter or shallow-gradient areas, approx 6 degrees, in order to maximize sediment retention. Effectiveness may be reduced on steeper slopes.

Sequence & Spacing Logistics

- **Spacing Interval:** Install dams **8 to 15 meters apart** (centre-to-centre), or spaced to maintain a maximum of **400mm vertical fall** between subsequent dams.
- **Catchment Strategy:** Stone dams should be paired with blocking of smaller upstream grips (≤1m) and bare peat revegetation to ensure full coverage of the catchment.

Material & Sizing Matrix

Dams are built using gritstone units (150mm – 450mm sizing). The number of dam units required depends on the width of the target gully:

Gully Width	Required Load / Units	Approximate Weight
< 2.0 meters wide	1 Unit	750 – 800 kg
2.0 to 3.0 meters wide	2 Units	1,500 – 1,600 kg
3.0+ meters wide	3 Units	2,250 – 2,400 kg

2. Construction Phase Steps

Step 1: Mapping & Pre-Planning

Draw up and finalise precise locations for individual dams prior to any on-site works. Refine these coordinates on the ground before structural deployment.

Step 2: Helicopter Delivery & Placement

Stone loads are delivered via helicopter. Units are dropped into the pre-planned pinch points, confluences, or mineral-to-peat transitions.

Step 3: Manual Reshaping & Profiling

After the helicopter drops, the stone dams should be manually shaped. This is ensures the dams meet the desired dimensions, profile angles, and geometric tolerances.

Step 4: Transverse & Longitudinal Dimensions

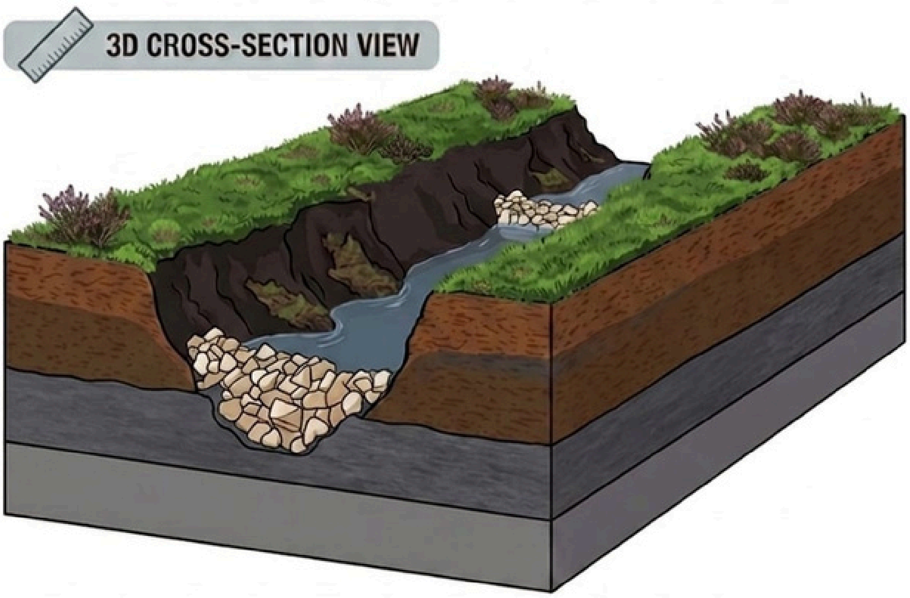
The dam should span the full cross-sectional width of the drain. The profile should be a minimum of **50cm high** and a maximum of 1.0m (for safety reasons), and should have a transverse width (upstream to downstream thickness downstream thickness) of at least **75cm**.

Step 5: Face Gradients & Slopes

The dam should have a steep **~60° face on the upstream side** to resist current forces, and a gradual, **~45° slope on the downstream face** to handle overflow energy.

Step 6: Central Spillway Notch Profiling

Structure the stone mound so it is distinctly higher on both outer sides than in the middle. This builds a central profile dip that funnels overflowing water down the centre face, preventing erosive side-scouring of the peat banks



Field Quality Checklist

- ☐ **Spaced Correctly:** 8m to 15m intervals or a maximum of 400mm vertical fall.
- ☐ **Height Compliance:** Min 50cm tall and strictly ≤ 1.0m maximum height limit.
- ☐ **Base Thickness:** Transverse width (upstream-to-downstream) is ≥ 75cm.
- ☐ **Channel Spanning:** Stone structure fully spans the entire width of the ditch.
- ☐ **Upstream Gradient:** Angled at approximately 60°.
- ☐ **Downstream Gradient:** Angled at approximately 45°.
- ☐ **Central Spillway:** The centre is lower than the outer edges to prevent side-scouring